Rohan Batra

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SUMMARY

Security engineer building autonomous vulnerability triage and remediation — Semgrep, Trivy, and Gitleaks in CI, OWASP Top 10 and STRIDE threat modelling, with human-in-the-loop guardrails and evals.

SKILLS

Security: Gitleaks, HashiCorp Vault, Semgrep, Trivy, age encryption, AppArmor, Burp Suite, Cerbos
Languages: Python, TypeScript, JavaScript, Shell, C, C++, Swift, TeX
Infrastructure: Docker, Helm, Nginx, Proxmox, Terraform, ADB, Evolution X, Fastboot, Garuda Linux, Kali NetHunter
Networking: DHCP / NAT, DNS, IP Allocation & Subnetting, pfSense, Pi-hole, NetBird
Backend: Flask, FastAPI, Pydantic, Uvicorn, Alembic, SQLAlchemy, Android SDK, Hugo, Microservices, REST APIs
DevOps: GitHub Actions, MkDocs, Obsidian, Termux, cargo-deny, cargo-llvm-cov, cargo-mutants
Data: MongoDB, Redis, PostgreSQL, ChromaDB, MySQL, DuckDB, DuckLake
AI/LLM: Claude, Anthropic API, LangChain, Ollama, Agent Evals & Guardrails, Agent Hooks, Agent Skills, Anthropic Messages API
Frontend: CSS, Next.js, React, Tailwind CSS, HTML, Dart

PROJECTS

Paridyum — Independent Multi-Tenant Platform *Rust, OpenTofu, Ansible, Talos Linux, Kubernetes, Cilium, Next.js, PostgreSQL*
Self-directed systems project: built solo from an empty directory to a running five-node Kubernetes cluster on self-hosted metal with a public internet edge.

- Built a **specification-as-software governance system**: 251 versioned artifacts (constitution, architecture decision records, standards, RFCs) with machine checks mapped outward to **NIST SSDF, CIS v8.1 and NIST CSF 2.0**, a multi-axis test-type taxonomy, and a precedence order in which existing code is the lowest authority.
- Delivered the full infrastructure path: hardened bare metal (AppArmor, nftables, confined QEMU/libvirt) through **Proxmox and OpenTofu** to **Talos Linux and Kubernetes** with Cilium as CNI and kube-proxy replacement, L2-announced load-balancer VIPs, and etcd snapshots that assert they captured something.
- Wrote **31,000+ lines of Rust** across two production tools — hexagonal architecture, trait-based abstraction, injectable clock adapters, and three independent SHA-256 hash chains for tamper-evident audit records.
- Engineered a **governable AI development harness**: a Claude Code plugin whose PreToolUse hooks make violating a locked architectural decision impossible rather than discouraged, with class-filtered context injection and append-only research provenance.

AutoTriage — Autonomous Vulnerability Triage & Remediation Agent *Python, Claude Agent SDK, Semgrep, Trivy, Gitleaks, Pydantic, Terraform*
An agent that triages real scanner findings end-to-end, with evals and human-in-the-loop guardrails.

- Built an autonomous vulnerability-triage agent on the Claude Agent SDK that ingests Semgrep, Trivy, and Gitleaks findings, reasons about severity and business impact, and files tickets, assigns owners, and drafts remediation PRs.
- Enforced human-in-the-loop guardrails via Pydantic validators — confidence-threshold escalation, fail-closed on malformed output, and prompt-injection defenses that treat scanner output as untrusted input.
- Built a fix-validation loop that applies proposed patches to an isolated copy of the code and re-runs the scanner, accepting a fix only when the finding is verifiably resolved with no regression.
- Designed an eval harness scoring triage decisions against a labelled ground-truth set: **100% verdict accuracy** across 44 live findings triaged into 24 tickets and 20 human escalations, backed by 68 tests at 89% coverage.

Gap Hunter — Offline Policy Gap Analyzer *Python, LangChain, Ollama, Pydantic, Electron, GitHub Actions*
HACKIITK 2026 finalist — top 0.4% of 1,300+ teams. Offline AI policy analysis against NIST CSF 2.0.

- Built an offline AI-powered policy gap analyzer mapping security policies against **NIST CSF 2.0** and the CIS MS-ISAC framework, producing evidence-grounded gap reports, revised policy drafts, and remediation roadmaps — with no external API dependency, so sensitive policy text never leaves the host.
- Designed a multi-stage pipeline (PDF/DOCX ingestion to hybrid retrieval combining vector, sparse, and reranking to gap analysis to policy revision) orchestrated via LangChain with Pydantic-validated structured outputs to ground LLM reasoning and reduce hallucination.
- Shipped a cross-platform Electron desktop app with local Ollama runtime integration and a **702-test suite** at 97.7% pass rate spanning unit, integration, property-based, and adversarial tests.
- Placed **top 0.4% of 1,300+ teams** and 9,000+ participants (C3iHub, IIT Kanpur Cybersecurity); one of 24 finalist teams invited to campus.

SWE-bench Task Audit *Python, Django, CI/CD*
An audit of SWE-bench Verified and a linter for patches that satisfy the grader without fixing the bug.

- Audited all **500 instances** of SWE-bench Verified and built a linter that detects patches satisfying the grader without fixing the underlying bug.

- Demonstrated a working exploit against the grading harness with a negative control, and made every claim independently reproducible: a proof workflow re-derives each result on a clean runner weekly and exits non-zero when reality diverges.

EXPERIENCE

Hardware & WireGuard Research Intern *2026-07 – Present*
Exaflair Technologies Pvt. Ltd.

Platform Engineering Intern *2025-11 – 2026-07*
Eve

- Founded the **Eve Gateway Stack** on APISIX and etcd — central routing, JWT authentication, rate limiting, CORS, and Prometheus metrics — establishing the platform's central trust and routing layer, with gateway-level tenant isolation, trusted tenant/user header injection, and Cerbos-based authorization rules.
- Implemented **schema-per-tenant multi-tenancy** using PostgreSQL search_path switching, with tenant context propagation and tenant-aware service and database isolation.
- Independently built **Eve-Testing**, a QA/SDET platform: automated API contract testing, a pytest harness with session client and test factories, orphan-sweep cleanup, matrix-driven test generation from spec spreadsheets, and product-gap tracking via xfail markers.

Technical Head *2025-09 – Present*
Google Developer Groups On Campus (GDGOC), UPES Dehradun

- Led and delivered technical workshops for **100+ participants** across cloud, system design, and full-stack development.
- Managed cross-functional teams executing production-grade initiatives.

Linux Support & Community Volunteer *2023-09 – 2025-04*
LinuxWale, Dehradun

- Onboarded **10+ students** to Linux — distribution selection, dual-boot setup, and driver/hardware troubleshooting across varied hardware.
- Wrote shell scripts to automate post-install configuration and routine setup tasks.

EDUCATION

UPES Dehradun — B.Tech (Hons.) Computer Science Engineering, Cyber Security *2023-08 – 2027-08 · CGPA 8.1/10*

ACHIEVEMENTS

- **Top 15% — TryHackMe cybersecurity learning platform**
- **Certificate of Appreciation — Golden Key Eagles Women Empowerment Center (Sep 2024)**
- **Certificate of Appreciation — HQ Uttarakhand Sub Area (Aug 2024)**
- **Certificate of Appreciation — Golden Key Tuskers, AWWA Week 2024 (Aug 2024)**
- **Semifinalist — Eureka! Junior 2022–23, E-Cell IIT Bombay (Issued Nov 2022; Credential ID: ecell2021)**
- **Semifinalist — Eureka! Junior 2021–22, E-Cell IIT Bombay (Issued Dec 2021; Credential ID: ecell2021)**